

Standard form



1

- | | | | | | | | |
|---|------------|---|------------|---|-----------|---|-----------|
| a | 2.49 | b | 20 000 000 | c | 41 300 | d | 0.009 |
| e | 897 000 | f | 50 300 | g | 3014 | h | 8 000 000 |
| i | 13 008 000 | j | 0.836 | k | 7 910 000 | l | 0.00714 |

2

- | | | | | | | | |
|---|---|---|---|---|----|---|-----------|
| a | 3 | b | 5 | c | -5 | d | 10^{-7} |
|---|---|---|---|---|----|---|-----------|

3

- | | | | | | | | |
|---|----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| a | 3×10^2 | b | 4.5×10^4 | c | 7×10^{-2} | d | 6.23×10^{-3} |
| e | 2×10^3 | f | 2×10^{-3} | g | 1×10^{-4} | h | 8.84×10^5 |
| i | 8.4626×10^7 | j | 6.14×10^0 | k | 3.47×10^{-4} | l | 3×10^{-5} |
| m | 7×10^{-2} | n | 7.64×10^{-1} | o | 2×10^6 | p | 1×10^{-4} |

4 D

5

- | | | | | | | | | |
|---|---|----------------------|----|-----|-----|----|----|-----|
| a | i | Yes | ii | No | iii | No | iv | Yes |
| | v | No | vi | Yes | | | | |
| b | | 4.5×10^{-4} | | | | | | |

6

- | | | | | | | | | | |
|---|---|-------------------|----|--------|---|---|--------------------|----|--------|
| a | i | 7.8×10^2 | ii | 10 | b | i | 4×10^{-3} | ii | 10 000 |
| c | i | 9×10^9 | ii | 30 000 | | | | | |

7



- a
- i 7.27×10^{-7} ii 2.22×10^{-7}
- b
- i 1.25×10^2 ii 1.25×10^{-6}
- c
- i 9.37×10^{-4} ii 6×10^{-9}
- d
- i 8.31×10^{-9} ii 5.13×10^{-9}
- e
- i 7.88×10^2 ii 2.93×10^{-2}
- f
- i 8.99×10^{-3} ii 1.99×10^{-6}
- g
- i $4.7 \times 10^9 \times 10^8$ ii $4.7 \times 10^{-8} \times 10^8$
- h
- i $2.12 \times 10^7 \times 10^2$ ii $2.12 \times 10^{-7} \times 10^4$

8

- a 6×10^{11} b 1.6×10^{-7} c 3×10^3 d 3×10^2
- e 4.5×10^4 f 2.1×10^{10} g 2.4×10^4 h 4.4×10^5

Standard form and calculators

9

- a 581 000 b 0.000316 c 2 500 000 d 0.00705

10

- a 0.003844 b 0.013

11

- a 1.120×10^{-15} b 3.342×10^2

- 12 At any particular time, approximately 2.32×10^5 aircraft are registered as being in flight over 7.772×10^{10} square metres of air space.



Significant figures

13

a Three

b Four

c Four

d Three

14

a 5.0

b 0.0060

c 801 600 000

d 0.0507

15

a

i 3

ii 422 m³

b

i 2

ii 2.5 mL

16

a 2.14×10^9

b 1.98×10^{-5}

17 4549

18 512

19

a 3499

b 2500

c 999

20

a 1

b 1.09

Applications

21 The diameter of a human cell

22

a 1.332×10^9

b 1.332×10^{21} litres

23

a 2 million each second

b 120 000 000

24

a 1 hundred thousandth of a metre

b 0.01 mm

25

a 3×10^{-1} km/s

b 3×10^5 km/s

c 1 000 000



26

a 2.5×10^5 km²

b 6.8×10^7

c 2.72×10^2 per km²

27

a 8×10^5 metres per hour

b 1000

28 3300

29 135 000 kg

30 2×10^7 g

31 7.95×10^7

32 2000 kg

33

a 1595 g

b 2×10^{26} atoms

34

a 2.107×10^{24} atoms

b 9 g

c 1.495×10^{-23} g

35

a 29 998 800 m

b 7×10^8 m

36 1.4130×10^9 days

37 1.25×10^{14}